Solutions to Second Problem Sheet
on
Analytical Mechanics
Autumn 2026

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Topic: Lagrangian Mechanics

"The reader will find no figures in this work. The methods which I set forth do not require either constructions or geometrical or mechanical reasonings: but only algebraic operations, subject to a regular and uniform rule of procedure."

-Joseph-Louis Lagrange (talking about his work Mécanique Analytique)

Problem 1

In the following cases, show that D'Alembert's principle holds and use this fact to find the equations of motion.

i) A block of mass m can slide freely on a wedge that is moving in a specified manner on a horizontal surface; that is, the coordinate X shown in Figure 1 is assumed to be a given function of time.

ii) Consider a bead of mass m that can slide along a rod which rotates about a vertical axis with constant angular velocity ω. The rod makes a constant angle θ_0 with the vertical axis (see Figure 2)

iii) A simple pendulum of mass m_2 hangs from a block of mass m_1 which is attached to a spring of stiffness k. The block lies on a horizontal surface without friction.

iv) A bead of mass m can slide freely along a hoop of radius a. The hoop rotates with constant angular velocity ω about a fixed point of the hoop on a horizontal plane (see Figure 4).

v) Two particles of mass m_1, m_2 are joined by a light rod of length l and move freely in space.

Solution 1

i) The position can be parametrised as

$$\boldsymbol{r} = (X + y\cot\theta_0)\hat{x} + y\hat{y}$$

and the constraint force on the mass is

$$\mathbf{F}^{(c)} = N(-\sin\theta_0\hat{x} + \cos\theta_0\hat{y})$$

so that it is readily verified that $\mathbf{F}^{(c)} \cdot \partial_y \boldsymbol{r} = 0$. For the equations of motion we have

$$(m\ddot{\boldsymbol{r}} - \mathbf{F}^{(a)}) \cdot \partial_y \boldsymbol{r} = 0$$

which are in turn equivalent to

$$\ddot{X}\cot\theta_0 + \ddot{y}(\cot^2\theta_0 + 1) + g = 0.$$

ii) Using spherical coordinates I will parametrise the position of the mass as $\boldsymbol{r} = r\hat{\rho}, \hat{\rho} = \cos\phi\hat{x} + \sin\phi\hat{y}$ where $\dot{\phi} = \omega$ and ω, θ_0 are constants. Given that the weight of the mass pulls it down, there

is a constraint force normal to the rod, meaning that

$$\mathbf{F}^{(c)} \cdot \frac{\partial \boldsymbol{r}}{\partial r} = \mathbf{F}^{(c)} \cdot \hat{\rho}$$

vanishes identically. To find the equations of motion we use the parametrisation $\boldsymbol{r} = r\sin\theta_0\cos\omega t\,\hat{x} + r\sin\theta_0\sin\omega t\,\hat{y} + r\cos\theta_0\hat{z}$ so that $(m\ddot{\boldsymbol{r}} - \mathbf{F}^{(a)}) \cdot \partial_r \boldsymbol{r} = 0$ is equivalent to

$$m(\ddot{r} - \omega^2 r\sin^2\theta_0) + mg\cos\theta_0 = 0.$$

iii) The position of the masses can be parametrised as (the y-axis is taken to be positive downwards)

$$\boldsymbol{r}_1 = x\hat{x}, \quad \boldsymbol{r}_2 = \boldsymbol{r}_1 + l(\sin\theta\hat{x} + \cos\theta\hat{y}),$$

and the constraint force on 1 is normal to the surface, i.e. $\mathbf{F}_1^{(c)} = -N\hat{y} + T(\sin\theta\hat{x} + \cos\theta\hat{y})$ and the one on the second mass is just $\mathbf{F}_2^{(c)} = -T(\sin\theta\hat{x} + \cos\theta\hat{y})$ so that for x

$$\mathbf{F}_1^{(c)} \cdot \partial_x \boldsymbol{r}_1 + \mathbf{F}_2^{(c)} \partial_x \boldsymbol{r}_2 = -N\hat{y}\cdot\hat{x} + T\sin\theta\hat{x}\cdot\hat{x} - T\sin\theta\hat{x}\cdot\hat{x} = 0$$

and $\partial_\theta \boldsymbol{r}_1 = 0$ so for θ we have

$$\mathbf{F}_2^{(c)} \cdot \partial_\theta \boldsymbol{r}_2 = -T(\sin\theta\hat{x} + \cos\theta\hat{y}) \cdot l(\cos\theta\hat{x} - \sin\theta\hat{y}) = -Tl(-\sin\theta\cos\theta + \sin\theta\cos\theta) = 0,$$

as usual, the equations of motion are recovered from the equations $\sum_\alpha (m_\alpha \mathbf{F}_\alpha^{(a)}) \cdot \partial_{q_i} \boldsymbol{r} = 0$

$$\begin{aligned} &x \to m_1\ddot{x} + kx + m_2\ddot{x} - m_2 l\ddot{\theta}\sin\theta = 0, \\ &\theta \to m_2\ddot{x}l\cos\theta - m_2 l\ddot{\theta}\sin\theta\cos\theta + l^2\ddot{\theta}\cos\theta\sin\theta + m_2 gl\sin\theta = 0. \end{aligned}$$

iv) The position of the mass can be parametrised by

$$\boldsymbol{r} = a([\cos\theta + \cos(\theta+\phi)]\hat{x} + [\sin\theta + \sin(\theta+\phi)]\hat{y})$$

where $\dot{\theta} = \omega$. The constraint force is a normal force produced by the hoop which is orthogonal to the vector $\partial_\phi \boldsymbol{r}$ since such a vector is tangent to the hoop. A straightforward calculation shows that the equation of motion reads

$$\ddot{\phi} + \omega^2\sin\phi = 0$$

v) Let $\boldsymbol{r}_1, \boldsymbol{r}_2$ be the position of the mass m_1, m_2, then define the vector

$$\boldsymbol{r}_1 = \mathbf{R} + \frac{m_2}{M} l\hat{n}, \quad \boldsymbol{r}_2 = \mathbf{R} - \frac{m_1}{M} l\hat{n}$$

where $\hat{n}$ is a unit vector in the direction of $\boldsymbol{r}_1 - \boldsymbol{r}_2$, then the constraint is given by $\mathbf{F}^{(c)} = N\hat{n}$ so that

$$\sum_\alpha \mathbf{F}^{(c)} \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} = N\hat{n} \cdot \frac{\partial \boldsymbol{r}_1}{\partial q_i} - N\hat{n} \cdot \frac{\partial \boldsymbol{r}_2}{\partial q_i} = N\hat{n} \cdot \frac{\partial \boldsymbol{r}_1 - \boldsymbol{r}_2}{\partial q_i} = \frac{N}{2\ell}\frac{\partial l^2}{\partial q_i} = 0$$

where $||\boldsymbol{r}_1 - \boldsymbol{r}_2|| = l$ is constant. The equations of motion depend on the applied forces. ✡

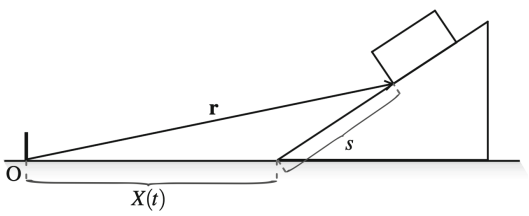


Figure 1: The position of the wedge is given by $X(t)$

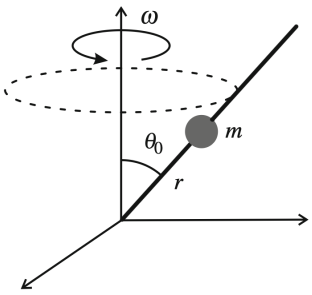


Figure 2: The bead can slide freely along the rod

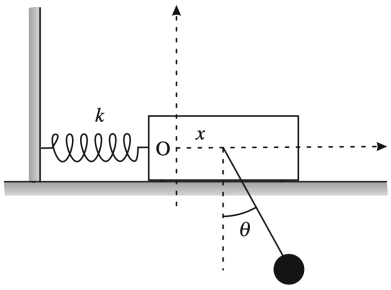


Figure 3: The bead can slide freely along the rod

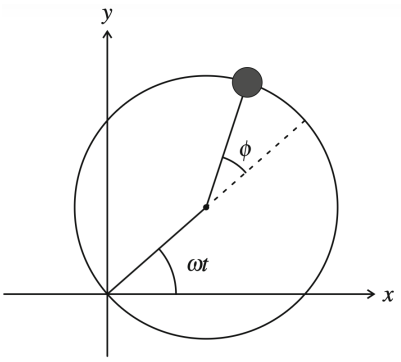


Figure 4: The bead can slide freely along the rod

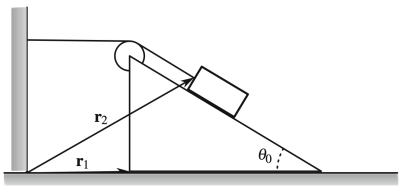


Figure 5: The bead can slide freely along the rod

Problem 2

Find the Lagrangian of each system in the previous problem.

Solution 2

i) $L(X, \dot{X}, s, \dot{s}) = \frac{1}{2}m[(\dot{X} + \dot{s}\cos\theta_0)^2 + \dot{s}\sin^2\theta_0] - mgs\sin\theta_0$

ii) $L(\dot{r}, r) = \frac{1}{2}m(\dot{r}^2 + \omega^2 r^2 \sin^2\theta_0) - \frac{1}{2}mgr^2\cos\theta_0$

iii) $L(x, \dot{x}, \theta, \dot{\theta}) = \frac{1}{2}(m_1 + m_2)\dot{x}^2 + mxl\dot{\theta}\cos\theta + \frac{1}{2}ml^2\dot{\theta}^2 - \frac{1}{2}kx^2 - mgl\cos\theta$

iv) $L(\phi, \dot{\phi}) = \frac{1}{2}m\left[a^2\omega^2 + a^2(\omega + \dot{\phi})^2 + 2a^2\omega(\omega + \dot{\phi})\right]$

v) $L(R, \dot{R}) = \frac{1}{2}(m_1 + m_2)\dot{R}^2$



Problem 3

If L is a Lagrangian for a system of n degrees of freedom satisfying Lagrange's equations, show that

$$L' = L + \frac{\mathrm{d}F(q_i, t)}{\mathrm{d}t}$$

also satisfies Lagrange's equations where F is any arbitrary, but differentiable, function of its arguments. This is sometimes known as a **gauge transformation**.

Solution 3

This is a clear consequence of the fact that the Euler-Lagrange equations are a necessary condition for the integral

$$I(C) = \int_{t_0}^{t_1} \mathrm{d}t\ L(q_i, \dot{q}_i, t)$$

to have an extreme value (when compared to neighbouring paths) but I will also show this by direct computation for pedagogical purposes.

Let me begin by computing the LHS of the Euler-Lagrange equations for L':

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L'}{\partial \dot{q}_i} - \frac{\partial L'}{\partial q_i} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i} + \frac{\partial F}{\partial q_j}\delta_{ij}\right) - \frac{\partial L}{\partial q_i} - \frac{\partial^2 F}{\partial q_i \partial q_j}\dot{q}_j - \frac{\partial^2 F}{\partial q_i \partial t}$$

$$= \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} + \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F}{\partial q_i} - \frac{\partial^2 F}{\partial q_i \partial q_j}\dot{q}_j - \frac{\partial^2 F}{\partial q_i \partial t},$$

and since

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F}{\partial q_i} = \frac{\partial^2 F}{\partial q_j \partial q_i}\dot{q}_j + \frac{\partial^2 F}{\partial t \partial q_i}$$

we appeal to Clairaut's theorem to show that, indeed,

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L'}{\partial \dot{q}_i} - \frac{\partial L'}{\partial q_i} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i}$$

which shows that

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L'}{\partial \dot{q}_i} - \frac{\partial L'}{\partial q_i} = 0 \iff \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0.$$



Problem 4

Show that the principle of virtual work

$$\sum_{\alpha=1}^{N} \mathbf{F}_\alpha^{(c)} \cdot \frac{\partial \boldsymbol{r}}{\partial q_i} = 0 \quad \text{for each } i = 1, \ldots, n$$

is independent of the set (q_i) of generalised coordinates chosen. More precisely, show that if (q_i') is

another minimal set, then the same principle holds but with q_i replaced by q_i'.

Solution 4

The desired result can be readily seen from

$$\sum_{\alpha=1}^{N} \mathbf{F}_\alpha^{(c)} \cdot \frac{\partial \boldsymbol{\imath}}{\partial q_i'} = \sum_{\alpha=1}^{N} \mathbf{F}_\alpha^{(c)} \cdot \frac{\partial \boldsymbol{\imath}}{\partial q_j} \frac{\partial q_j}{\partial q_i'} = \frac{\partial q_j}{\partial q_i'} \left(\sum_{\alpha=1}^{N} \mathbf{F}_\alpha^{(c)} \cdot \frac{\partial \boldsymbol{\imath}}{\partial q_j} \right).$$



Problem 5

Show that Lagrange's equations can also be written as

$$\frac{\partial \dot{T}}{\partial \dot{q}_j} - 2\frac{\partial T}{\partial q_j} = Q_j.$$

These are sometimes known as the **Nielsen** form of Lagrange's equations

Solution 5

Lagrange's equations read

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_j} - \frac{\partial T}{\partial q_j} = Q_j$$

which on expanding we get

$$\frac{\partial}{\partial q_i}\frac{\partial T}{\partial \dot{q}_j} \cdot \dot{q}_i + \frac{\partial}{\partial \dot{q}_i}\frac{\partial T}{\partial \dot{q}_j} \cdot \ddot{q}_i + \frac{\partial}{\partial t}\frac{\partial T}{\partial \dot{q}_j} - \frac{\partial T}{\partial q_j} = Q_j \qquad (*)$$

so that changing the order of differentiation we get that the first term is

$$\frac{\partial}{\partial \dot{q}_j}\frac{\partial T}{\partial q_i} \cdot \dot{q}_i = \frac{\partial}{\partial \dot{q}_j}\left(\frac{\partial T}{\partial q_i}\dot{q}_i\right) - \frac{\partial T}{\partial q_i}\frac{\partial \dot{q}_i}{\partial \dot{q}_j} = \frac{\partial}{\partial \dot{q}_j}\left(\frac{\partial T}{\partial q_i}\dot{q}_i\right) - \frac{\partial T}{\partial q_i}\delta_{ij}$$

and the second term can be written as

$$\frac{\partial}{\partial \dot{q}_j}\left(\frac{\partial T}{\partial \dot{q}_i}\ddot{q}_i\right)$$

and this way $(*)$ becomes

$$\frac{\partial}{\partial \dot{q}_j}\left(\frac{\partial T}{\partial q_i}\dot{q}_i + \frac{\partial T}{\partial \dot{q}_i}\ddot{q}_i + \frac{\partial T}{\partial t}\right) - \frac{\partial T}{\partial q_j} - \frac{\partial T}{\partial q_j} = Q_j$$

and the conclusion now readily follows.



Problem 6

Show that the Lagrangian

$$L = \frac{1}{2}m\mathbf{v}^2 + e\mathbf{A}\cdot\mathbf{v} - e\varphi$$

gives rise to the **Lorentz equation** for a particle of charge e

$$\mathbf{F} = e\left(\mathbf{E} + \mathbf{v}\times\mathbf{B}\right)$$

through the Euler-Lagrange equations.

Solution 6

Write the Lagrangian as

$$L = \frac{1}{2}mv_jv_j + eA_jv_j - e\varphi$$

then the Euler-Lagrange equations read

$$\frac{\partial L}{\partial x_i} = e\frac{\partial A_j}{\partial x_\sigma}\frac{\mathrm{d}x_\sigma}{\mathrm{d}x_i}v_j - e\frac{\partial\varphi}{\partial x_i} = e\frac{\partial A_j}{\partial x_i}v_j - e\frac{\partial\varphi}{\partial x_i}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial v_i} = \frac{\mathrm{d}}{\mathrm{d}t}\left(m\frac{\mathrm{d}v_j}{\mathrm{d}v_i}v_j + eA_j\frac{\mathrm{d}v_j}{\mathrm{d}v_i}\right) = m\frac{\mathrm{d}v_i}{\mathrm{d}t} + e\frac{\mathrm{d}A_i}{\mathrm{d}t} = ma_i + e\frac{\partial A_i}{\partial x_l}v_l + e\frac{\partial A_i}{\partial t}$$

$$\frac{\partial L}{\partial x_i} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial v_i} = 0 \iff ma_i + e\frac{\partial A_i}{\partial x_l}v_l + e\frac{\partial A_i}{\partial t} = e\frac{\partial A_j}{\partial x_i}v_j - e\frac{\partial\varphi}{\partial x_i}$$

or, equivalently,

$$ma_i = e\left(\frac{\partial A_j}{\partial x_i}v_j - \frac{\partial\varphi}{\partial x_i} - \frac{\partial A_i}{\partial x_l}v_l - \frac{\partial A_i}{\partial t}\right)$$

and the Lorentz equation (using $\mathbf{E} = -\nabla\varphi - \partial_t\mathbf{A}$ and $\mathbf{B} = \nabla\times\mathbf{A}$) is

$$ma_i = e\left(-\frac{\partial\varphi}{\partial x_i} - \frac{\partial A_i}{\partial t} + \varepsilon_{ijk}\varepsilon_{klm}v_j\partial_l A_m\right) = e\left(-\frac{\partial\varphi}{\partial x_i} - \frac{\partial A_i}{\partial t} + v_j\frac{\partial A_j}{\partial x_i} - v_j\frac{\partial A_i}{\partial x_j}\right)$$

which is clearly equivalent to the previous result.



Problem 7

The electromagnetic field is invariant under a Gauge transformation of the scalar and vector potential given by

$$\mathbf{A} \to \mathbf{A} + \nabla\psi(\mathbf{r},t)$$
$$\varphi \to \varphi - \frac{\partial\psi}{\partial t},$$

where ψ is any differentiable function. What effect does this gauge transformation have on the Lagrangian of a particle moving in the electromagnetic field? Is the motion affected?

Solution 7

The effect of such a transformation on the Lagrangian is

$$L \to L + e\nabla\psi(\mathbf{r},t)\cdot\mathbf{v} + e\frac{\partial\psi(\mathbf{r},t)}{\partial t} = e\frac{\partial\psi(\mathbf{r},t)}{\partial r_j}\frac{\mathrm{d}r_j}{\mathrm{d}t} + e\frac{\partial\psi(\mathbf{r},t)}{\partial t} = e\frac{\mathrm{d}\psi(\mathbf{r},t)}{\mathrm{d}t}$$

and this is where the "gauge transformation" name has a connection with **Problem 3**. We already

know that the such a transformation does not affect the motion, the Euler-Lagrange equations are equivalent.



Problem 8

Let $q_1, \ldots, q_n$ be a set of independent generalised coordinates for a system of n degrees of freedom, with a Lagrangian $L(q, \dot{q}, t)$. Suppose we transform to another set of independent coordinates $s_1, \ldots, s_n$ by means of transformation equations

$$q_i = q_i(s_1, \ldots, s_n; t), \quad i = 1, \ldots, n.$$

(Such a transformation is called a **point transformation**.) Show that if the Lagrangian function is expressed as a function L' of $s_j, \dot{s}_j$, and t through the transformation equations, then L' satisfies Lagrange's equations with respect to the s coordinates:

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L'}{\partial \dot{s}_j} - \frac{\partial L'}{\partial s_j} = 0.$$

Solution 8

Similar to **Problem 3**, this is a direct consequence of the fact that the Euler-Lagrange equations are a necessary condition for an integral of the form

$$I(C) = \int_{t_0}^{t_1} \mathrm{d}t \; L(q_j, \dot{q}_j, t)$$

to have an extreme value (in the sense explained in class), so if $L'(s_j, \dot{s}_j, t)$ is the Lagrangian L expressed as a function of $s_j, \dot{s}_j, t$, then clearly,

$$I(C) = \int_{t_0}^{t_1} \mathrm{d}t \; L'(s_j, \dot{s}_j, t)$$

which leads to the set of equations

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L'}{\partial \dot{s}_j} - \frac{\partial L'}{\partial s_j} = 0.$$



Problem 9

A Lagrangian for a particular physical system can be written as

$$L' = \frac{1}{2}m\left(a\dot{x}^2 + 2b\dot{x}\dot{y} + c\dot{y}^2\right) - \frac{1}{2}K\left(ax^2 + 2bxy + cy^2\right),$$

where a, b, and c are arbitrary constants but subject to the condition that $b^2 - ac \neq 0$. Find the equations of motion and examine particularly the two cases $a = 0 = c$ and $b = 0, c = -a$. What is the physical system described by the above Lagrangian? Show that the usual Lagrangian for this system is related to L' by a point transformation (cf. **Problem 8**). What is the significance of the condition on the value of $b^2 - ac$?

Solution 9

The equations of motion for x, y read

$$x: \; ma\ddot{x} + mb\ddot{y} = -kax - kby, \quad y: \; mc\ddot{y} + mb\ddot{x} = -kcy - kbx$$

and the particular case $a = 0 = c$ yields

$$x: \ mb\ddot{y} = -kby \quad y: \ mb\ddot{x} = -kbx$$

so the system is a set of two independent harmonic oscillators of the same angular frequency $\omega = \sqrt{k/m}$. The case $b = 0, c = -a$ yields

$$x: \ mc\ddot{x} = -kcx, \quad y: \ mc\ddot{y} = -kcy$$

which is the same system as before. The system is a particle subject to a force given by

$$\mathbf{F} = -K\mathsf{M}\boldsymbol{r}, \quad \mathsf{M} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad \boldsymbol{r} = \begin{pmatrix} x \\ y \end{pmatrix}.$$

The point transformation relating this to the usual Lagrangian for this problem is then clearly

$$q_1(x, y) = ax + by, \quad q_2(x, y) = bx + cy.$$

The condition on the value of $b^2 - ac$ amounts to say that the matrix M is invertible, which in the context of linear algebra means that the system cannot be reduced to two one-degree of freedom problems, although this can also be seen from the fact that if $b^2 - ac = 0$, then one could write

$$x' = \sqrt{a}x, \quad y' = \sqrt{c}y$$

to get

$$L' = \frac{1}{2}m(\dot{x}'^2 + \dot{y}'^2) - \frac{1}{2}K(x'^2 + y'^2)$$

which would be a system of two independent differential equations.

Problem 10

The *Emden-Fowler equation*

$$\ddot{q} + \frac{2}{t}\dot{q} + q^k = 0$$

where k is a constant, arises in the study of the spherically symmetric equilibrium configurations of a self-gravitating polytropic fluid; t is proportional to the radial distance and q is related to the density of the fluid. Find a Lagrangian for such an equation.

Solution 10

$$L = \frac{t^2\dot{q}^2}{2} - \frac{t^2 q^{k+1}}{k+1}.$$



Problem 11

Find a Lagrangian for the *Poisson-Boltzmann equation*

$$\ddot{q} = -\frac{k}{t}\dot{q} - ae^q,$$

where k and a are constants restricted by the conditions $k \neq 0$ and $a \in \{-1, 1\}$.

Solution 11

$$L = \frac{1}{2}t^k\dot{q}^2 - at^k e^q$$



Problem 12

Find the Lagrangian for a particle subject to a frictional force proportional to the velocity,

$$\ddot{q} = -2\gamma\dot{q},$$

where γ is a constant.

Solution 12

$$L = m(\dot{q}\ln\dot{q} - \dot{q}) - 2m\gamma q.$$



Problem 13

Find a Lagrangian for a system of variable mass in a uniform gravitational field whose equation of motion is given by

$$\ddot{q} = -g - u\frac{\mathrm{d}\ln m}{\mathrm{d}t}.$$

Solution 13

$$L = \frac{1}{2}\dot{q}^2 - \left(g + u\frac{\mathrm{d}\ln m}{\mathrm{d}dt}\right)q.$$



Problem 14

Consider the Lagrangian of a particle moving in a plane under the influence of a central force with potential V

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - V\left(\sqrt{x^2+y^2}\right).$$

i) Show that

$$x^{(s)} = x\cos s - y\sin s, \quad y^{(s)} = x\sin s + y\cos s, \quad t^{(s)} = t, \quad s \in \mathbb{R}$$

define a one-parameter family of strict symmetries of the Lagrangian.

ii) Directly verify that the functions η_i, ξ satisfy

$$\frac{\partial L}{\partial q_i}\eta_i + \frac{\partial L}{\partial \dot{q}_i}\dot{\eta}_i + \frac{\partial L}{\partial t}\xi + \dot{\xi}\left(L - \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right) = 0.$$

iii) Find the constant of the motion associated to the symmetry. Does it have a clear meaning?

Furthermore, discuss the same problem for a particle moving in space, define three one-parameter families of strict symmetries associated to rotations and deduce the three conserved quantities.

Solution 14

i) Recall that the family of transformations define a strict symmetry of the Lagrangian if

$$\int_{\delta t} \mathrm{d}t \left[L(q_i, \mathrm{d}q_i/\mathrm{d}t, t) - \frac{\mathrm{d}t^{(s)}}{\mathrm{d}t} L\left(q_i^{(s)}, \frac{\mathrm{d}q_i/\mathrm{d}t}{\mathrm{d}t^{(s)}/\mathrm{d}t}, t^{(s)} \right) \right] = 0$$

which can be readily seen to be satisfied from the fact that

$$\frac{\mathrm{d}t^{(s)}}{\mathrm{d}t} = 1, \quad \text{and} \quad \left(x^{(s)}\right)^2 + \left(y^{(s)}\right)^2 = x^2 + y^2.$$

ii) The functions η_i, ξ are defined by

$$\eta_i = \left. \frac{\partial q_i^{(s)}}{\partial s} \right|_{s=0}, \quad \xi = \left. \frac{\partial t^{(s)}}{\partial s} \right|_{s=0}$$

and a straightforward calculation yields $\eta_1 = -y, \eta_2 = x, \xi = 0$ and hence

$$\frac{\partial L}{\partial q_i}\eta_i + \frac{\partial L}{\partial \dot{q}_i}\dot{\eta}_i + \frac{\partial L}{\partial t}\xi + \dot{\xi}\left(L - \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i \right) = -\frac{\partial L}{\partial x}y + \frac{\partial L}{\partial y}x - \frac{\partial L}{\partial \dot{x}}\dot{y} + \frac{\partial L}{\partial \dot{y}}\dot{x}$$

which by writing $r = \sqrt{x^2 + y^2 + z^2}$ one can see how

$$= -\frac{\mathrm{d}V}{\mathrm{d}r}\frac{xy}{r} + \frac{\mathrm{d}V}{\mathrm{d}r}\frac{yx}{r} - m\dot{x}\dot{y} + m\dot{y}\dot{x} = 0.$$

iii) The constant of the motions is now given by

$$N = \frac{\partial L}{\partial \dot{q}_i}\eta_i = -m\dot{x}y + m\dot{y}x$$

because $\xi = 0$. This can be easily identified with the z-component of the angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, which is to be expected since the one-parameter family of transformations define a rotation in the plane (about the z-axis).

For the case of a particle in space one can proceed similarly, defining

$$y^{(\vartheta)} = y\cos\vartheta - z\sin\vartheta, \quad z^{(\vartheta)} = y\sin\vartheta + z\cos\vartheta, \quad t^{(s)} = t,$$

and

$$x^{(\phi)} = x\cos\phi + z\sin\phi, \quad z^{(\phi)} = -x\sin\phi + z\cos\phi, \quad t^{(s)} = t,$$

as the two additional families of strict symmetries (the first one being the same as in the case of movement in the plane) and the associated integrals of the motion can be readily seen to be the x and z components of the angular momentum, respectively.

Problem 15

Find the variational symmetries of the Lagrangian

$$L = \frac{1}{2}m(\dot{r}^2 + \omega^2 r^2 \sin^2\theta_0) - mgr\cos\theta_0,$$

where m, ω, g, and θ_0 are constants. Show that there are two one-parameter families of variational symmetries with $\xi = 0$.

Solution 15

The functions η_i, ξ are determined by the following PDE

$$\frac{\partial L}{\partial q_i}\eta_i + \frac{\partial L}{\partial \dot{q}_i}\left(\frac{\mathrm{d}\eta_i}{\mathrm{d}t} - \dot{q}_i\frac{\mathrm{d}\xi}{\mathrm{d}t}\right) + \frac{\partial L}{\partial t}\xi + L\frac{\mathrm{d}\xi}{\mathrm{d}t} = \frac{\mathrm{d}G}{\mathrm{d}t},$$

where we have only one degree of freedom so

$$\frac{\partial L}{\partial r} = m\omega^2 r\sin^2\theta_0 - mg\cos\theta_0, \quad \frac{\partial L}{\partial \dot{r}} = m\dot{r},$$

and thus the PDE reads

$$\begin{aligned}\left(m\omega^2 r\sin^2\theta_0 - mg\cos\theta_0\right)\eta + m\dot{r}\left(\frac{\partial\eta}{\partial r}\dot{r} + \frac{\partial\eta}{\partial t} - \dot{r}^2\frac{\partial\xi}{\partial r} - \dot{r}\frac{\partial\xi}{\partial t}\right)\\ + \left(\frac{1}{2}m(\dot{r}^2 + \omega^2 r^2\sin^2\theta_0) - mgr\cos\theta_0\right)\left(\frac{\partial\xi}{\partial r}\dot{r} + \frac{\partial\xi}{\partial t}\right) = \frac{\partial G}{\partial r}\dot{r} + \frac{\partial G}{\partial t},\end{aligned}$$

and since the functions η, ξ, G depend only on r and t, then we can equate the powers of $\dot{r}$ to get

$$\dot{r}^3\left[-\frac{1}{2}m\frac{\partial\xi}{\partial r}\right] = 0 \tag{1}$$

$$\dot{r}^2\left[m\frac{\partial\eta}{\partial r} - \frac{1}{2}m\frac{\partial\xi}{\partial t}\right] = 0 \tag{2}$$

$$\dot{r}\left[m\frac{\partial\eta}{\partial t} + \left(\frac{1}{2}m\omega^2 r^2\sin^2\theta_0 - mgr\cos\theta_0\right)\frac{\partial\xi}{\partial r}\right] = \dot{r}\left[\frac{\partial G}{\partial r}\right] \tag{3}$$

$$\left(m\omega^2 r\sin^2\theta_0 - mg\cos\theta_0\right)\eta + \left(\frac{1}{2}m\omega^2 r^2\sin^2\theta_0 - mgr\cos\theta_0\right)\frac{\partial\xi}{\partial t} = \frac{\partial G}{\partial t} \tag{4}$$

whence (1) yields $\partial\xi/\partial r = 0$ so that $\xi = A(t)$ and then (3) becomes

$$\frac{\partial G}{\partial r} = m\frac{\partial\eta}{\partial t},$$

but from (2) one has

$$\frac{\partial\eta}{\partial r} = \frac{1}{2}\frac{\partial\xi}{\partial t} \to \frac{\partial\eta}{\partial r} = \frac{1}{2}\frac{\mathrm{d}A}{\mathrm{d}t},$$

meaning that

$$\eta = B_1(t)r + B_2(t), \quad B_1 = \frac{1}{2}\frac{\mathrm{d}A}{\mathrm{d}t}$$

and now we can equate the mixed partial derivatives $\partial_r\partial_t, \partial_t\partial_r$ of G to get (using (4) and $\eta = B_1 r + B_2$)

$$\begin{aligned}m\frac{\mathrm{d}^2 B_1}{\mathrm{d}t^2}r + m\frac{\mathrm{d}^2 B_2}{\mathrm{d}t^2} = m\omega^2\sin^2\theta_0(B_1 r + B_2) + \left(m\omega^2 r\sin^2\theta_0 - mg\cos\theta_0\right)B_1\\ + \left(m\omega^2 r\sin^2\theta_0 - mg\cos\theta_0\right)\frac{\mathrm{d}A}{\mathrm{d}t}\end{aligned}$$

and since A, B_1, B_2 all depend on t only we can equate the powers of r to get

$$\frac{\mathrm{d}^2 B_1}{\mathrm{d}t^2} = 2\omega^2\sin^2\theta_0 B_1 + \omega^2\sin^2\theta_0\frac{\mathrm{d}A}{\mathrm{d}t} \iff \frac{\mathrm{d}^2 B_1}{\mathrm{d}t^2} = \omega^2\sin^2\theta_0\left(2B_1 + \frac{\mathrm{d}A}{\mathrm{d}t}\right) \tag{$*$}$$

$$\frac{\mathrm{d}^2 B_2}{\mathrm{d}t^2} = \omega^2\sin^2\theta_0 B_2 - g\cos\theta_0\frac{\mathrm{d}A}{\mathrm{d}t} \tag{$**$}$$

and on using $2B_1 = \mathrm{d}A/\mathrm{d}t$ we get from $(*)$:

$$\frac{\mathrm{d}^2 B_1}{\mathrm{d}t^2} = \Omega^2 B_1, \quad \Omega^2 = 4\omega^2 \sin^2\theta_0$$

whose solution is $B_1 = Ce^{\Omega t} + De^{-\Omega t}$ and thus

$$\xi = A(t) = Ee^{\Omega t} - Fe^{-\Omega t}, \quad E = \frac{2C}{\Omega}, \quad F = \frac{2D}{\Omega}$$

then by solving

$$\frac{\partial t^{(s)}}{\partial s} = Ee^{\Omega t^{(s)}} - Fe^{-\Omega t^{(s)}}$$

one could, in principle, get $t^{(s)}$ and similarly, substituting the given expression for A into $(**)$ one could find B_2 by solving the ODE, notice that the solution of A contains an arbitrary constant and the solution for B_1, B_2 contain two arbitrary constants each, meaning that this Lagrangian possesses five variational symmetries, in particular, for $\xi = 0$ one has $B_1 = 0$ and from $(**)$ we also have

$$B_2 = ae^{ft} + be^{-ft}, \quad f = \omega \sin\theta_0$$

and hence we can use (4) and the fact that for $B_1 = 0$ one has $\eta = B_2$ to get G:

$$\frac{\partial G}{\partial t} = (m\omega r \sin\theta_0 - mg\cos\theta_0)B_2 \implies G = \left(m\omega^2 r \sin^2\theta_0 - \frac{mg\cos\theta_0}{\omega\sin\theta_0}\right)\left(ae^{\omega\sin\theta_0 t} + be^{-\omega\sin\theta_0 t}\right),$$

the constant of the motion requires us only to know ξ, η, G so in this case we can see that the integral of the motion is simply

$$N = \frac{\partial L}{\partial \dot{r}}\eta - G = m\dot{r}\eta - G$$

$$= m\dot{r}(ae^{\omega\sin\theta_0 t} + be^{-\omega\sin\theta_0 t}) - \left(m\omega^2 r\sin^2\theta_0 - \frac{mg\cos\theta_0}{\omega\sin\theta_0}\right)\left(ae^{\omega\sin\theta_0 t} + be^{-\omega\sin\theta_0 t}\right)$$

and now, given the definitions of ξ, η one requires

$$\left.\frac{\partial t^{(s)}}{\partial s}\right|_{s=0} = 0, \quad \text{and} \quad \left.\frac{\partial r^{(s)}}{\partial s}\right|_{s=0} = \left[ae^{ft^{(s)}} + be^{-ft^{(s)}}\right]_{s=0} \quad \forall s$$

since $\xi = 0$, we get $t^{(s)} = t^{(0)}$ and for $r^{(s)}$ we have

$$r^{(s)} = r^{(0)} + ase^{ft^{(0)}} + bse^{-ft^{(0)}}.$$



Problem 16

Consider a particle of mass m subject to a central force, we showed in class that such a force is always derivable from a potential $V(r)$, in particular, using spherical coordinates the Lagrangian takes the form

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2) - V(r)$$

which leads to the fact that $p_\phi \equiv \partial L/\partial\dot{\phi}$ is a conserved quantity. Using the definition of angular momentum $\mathbf{L} = \mathbf{r}\times m\dot{\mathbf{r}}$, show that $p_\phi = L_z$, i.e., that the conserved quantity is the zth component of the angular momentum in Cartesian coordinates.

Solution 16

The proof is a straightforward computation, using $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$ we have

$$\begin{aligned}\frac{L_z}{m} = (x\dot{y} - y\dot{x}) = r\dot{r}\sin\theta\cos\phi\sin\phi\sin\theta + r^2\dot{\phi}\sin\theta\cos\phi\cos\phi\sin\theta \\ + r^2\dot{\theta}\sin\theta\cos\phi\sin\phi\cos\theta - r\dot{r}\sin\theta\sin\phi\cos\phi\sin\theta \\ - r^2\dot{\theta}\sin\theta\sin\phi\cos\phi\cos\theta + r^2\dot{\phi}\sin\theta\sin\phi\sin\phi\sin\theta\end{aligned}$$

and clearly, the terms containing $r\dot{r}$ and $r^2\dot{\theta}$ cancel each other out, so we are only left with the terms containing $r^2\dot{\phi}$,

$$L_z = mr^2\dot{\phi}\left(\sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi\right) = mr^2\dot{\phi}\sin^2\theta,$$

which by direct differentiation one can see that this is $\partial L/\partial\dot{\phi}$.

